ClimateShield 360: Karachi Extreme Weather & Crisis Platform

HackInsight 2026: Data-to-Decision • Climate & Infrastructure Analytics Track

Deliverable: Summary.pdf (1-Page)

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1. THE PROBLEM & PRIMARY DECISION MAKER

Context: Karachi (17M+ citizens) suffers severe monsoon flash-flooding in the Lyari/Malir basins, submerging underpasses and arterial corridors under 2–4 feet of water, coupled with lethal urban heatwaves.

Primary User: Emergency Municipal Dispatchers at **PDMA Sindh / KMC** and **Karachi Traffic Police (KTP)**.

Actionable Decision: Provides a **2 to 4-hour tactical advance window** to pre-deploy dewatering suction pumps and position diversion cones *before* arterial highways choke, replacing delayed post-facto reactions.

2. MULTI-SOURCE WAREHOUSE & DATA INTEGRITY

Volume & Span: 592,176 hourly observations (2021–2026) across 12 Pakistani cities (exceeds 50k threshold by 11x).

- **Independent Sources:** Open-Meteo (Precipitation/Heat), Copernicus GloFAS (River Runoff), NDMA/PMD Bulletins, PBS 2023 Census (Density), OpenStreetMap (Roads & Hospitals).
- **Reproducibility:** Rebuilt via `make data` from raw & cleaned layers into gold DuckDB (`climateshield.duckdb`) with SHA-256 cryptographic manifest.

3. DATA QUALITY AGENT (PANDERA FRAMEWORK)

Automated Decision Engine: Replaces naive dropouts with explicit rule-based decisions audited in `quality_log`:

- **FIX:** Standardizes UTC to PKT timezones, converts rainfall/temperature units, and resolves city name variants.
- **FILL:** Imputes short sensor gaps (≤3h) via local spline; flags imputed rows.
- **FLAG:** Quarantines impossible sensor spikes (>150mm/h) and unverified reports without dropping raw evidence.

4. MODEL VS DECLARED BASELINE (DELIVERABLE 3)

Declared Baseline: 24-hour persistence ($ŷ_t = y_{t-24 ext{h}}$).

Model: LightGBM Quantile Regressors (Q_{10}, Q_{50}, Q_{90}) trained on 403,188 rows (2021–2024, covers 2022 super floods); tested on 189,168 unseen rows (2025–2026).

Evaluation Metric	24h Baseline	LightGBM Model	Improvement
Test Rainfall MAE	0.1055 mm	**0.0685 mm**	**+35.1% Lower MAE**
Prediction Interval Coverage	Point Only	**97.42%**	Calibrated ($Q_{10} o Q_{90}$)

Fairness by Group: Audited across 12 cities; weakest group identified with operational dispatcher guidance.

5. INSIGHT AGENT (RAG) & 8 IMPACT DOMAINS

Strict Responsible AI Protocol:

- **Zero Synthetic Numbers:** Every metric in briefs originates directly from DuckDB gold tables with row IDs cited.
- **Evidence Fallback:** Automatically returns *"I don't know"* when queries lack supporting official documentation.
- **8 Impact Sectors:** Physical, Transport, Health, Energy, Economic, Agriculture, Social, Emergency Management.

6. PIPELINE AUTOMATION & BUSINESS MODEL

Hourly Pipeline: Live ingestion daemon fetches delta feeds hourly; tracks freshness radar and logs unedited `prefect_runs.csv`.

Interactive Stack: FastAPI async backend (port 8002) + Vite React SPA (port 5173) with resilient CARTO Dark 3D map.

Business Sustainability: Public-Private-Civic Partnership (Sindh Resilience Project / KMC) with low-overhead operating footprint of **~$30–$65 / month**.

HackInsight 2026 Submission Gate: All 5 Deliverables Complete • 82/82 Pytest Passed • Rebuild: `make data` • Status: LIVE

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